Problem Set 1 (two pages)

Statistics 24510-30040 (Winter 2025)

Due 9 am Tuesday, January 14, 2025.

Requirements

- Put the assignment number, course number (24510 or 30040) and your name at the top of the first page.
- Type (preferred) or neatly handwrite on standard letter-size papers (hardcopy or virtual) using dark ink with light background.
- Scan and save your work as a letter-size pdf file named "Lastname-Firstname-hw1.pdf" (abbreviations allowed for long names).
- Upload your pdf to Gradescope under Pset1. Be sure to mark the pages corresponding to each problem during submission.
- Show your work. Provide detailed derivations and arguments in order to receive credit.
- You may discuss methods with others, but you should devise and write your answers by yourself completely and independently.

Problem assignments (This warmup problem set is mostly a review, relevant to contents in chapters 3, 4, 8 and 9 in Rice's book.)

1. (*Normal samples*)

   Suppose $X_1, \cdots, X_n$ are *i.i.d.* $\sim N(\mu, 2^2)$ (that is, X_i's are independent with the common normal distribution).

   (a) What is the distribution of the sample mean $\bar{X} = \sum_{i=1}^{n} X_i/n$ (which is an unbiased estimator of μ)?

   (b) Provide a 95% confidence interval for μ, based on the distribution in Part (a).

   (c) To make the 95% confidence interval for μ with length at most 0.4, what is the smallest sample size n needed?

2. (*Binomial sample size*)

   Assume $X \sim Binomial(n, p)$. Then $\hat{p} = X/n$ is the maximum likelihood estimator (MLE) of p.

   The endpoints of the 95% Wald asymptotic confidence interval for p has the form $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$, which can be used in the following.

   (a) Find the smallest integer n_0 such that for any sample size $n \geq n_0$, The length of the 95% confidence interval is ≤ 0.2, regardless of the value of $\hat{p} \in [0, 1]$.

   (b) Based on the above derivation, if we know n^* is the smallest sample size to guarantee the confidence interval for p to be of length $\leq L$,

      i. What is the minimal sample size to guarantee a shorter 95% confidence interval of length $\leq L/2$?

      ii. And, what is the minimal sample size to guarantee a 95% confidence interval of length $\leq L/10$?

3. *(Poisson distribution and the moment generating function)*

   Assume $X \sim Poisson(\lambda)$.

   (a) Derive the moment generating function $M(t) = \mathbb{E}(e^{tX})$. You need to evaluate the sum of an infinite series.

   (b) Calculate the third derivative $M^{(3)}(t) = \frac{d^3}{dt^3} M(t)$.

   (c) Using the moment generating function property $M^{(k)}(0) = \mathbb{E}(X^k)$ to derive $\mathbb{E}(X^4)$.

4. (*Exponential distribution*)

   Suppose $X \sim Exp(\lambda)$ with probability density function $f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0; \\ 0, & x < 0. \end{cases}$

   Derive the mean $\mathbb{E}(X)$ and variance $Var(X)$. Show the integrals used in your derivation.

5. (*Distributions of the sum*)

   Suppose X and Y are independent random variable with probability density functions $f_X(x)$ and $f_Y(y)$, respectively. Let $W = X + Y$.

   (a) Derive the distribution function $F_W(w) = \mathbb{P}(W \leq w)$ in terms of integrals of f_X and f_Y.

   (b) Derive the density function $f_W(w) = \frac{d}{dw} F_W(w)$ in terms of an integral of f_X and f_Y.
   (The integral is called the convolution of functions f_X and f_Y.)

6. (*Exponential and Gamma distributions, MLE, mathematical induction*)

   Suppose $X_1, \cdots, X_n$ are i.i.d. $Exp(\lambda)$ with probability density function $f(x) = \lambda e^{-\lambda x}, x \geq 0$.

   (a) Derive the maximum likelihood estimator (MLE) for the parameter λ.
      i. Write the likelihood function given the data $X_1, \cdots, X_n$.
      ii. Write the log-likelihood function.
      iii. Use differentiation to derive the MLE for λ.

   (b) Derive the probability density function for the sum $S_n = \sum_{i=1}^{n} X_i$.
   Start from $n = 2$ using the density for the sum of independent random variables derived earlier.
   Then carry out a mathematical induction on n.

   The sums are of gamma distribution with appropriate parameters. Show your derivation steps.